
Problem Set 9

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Problem 1

- hi

$$\det \begin{pmatrix} 1 & z_1 \\ 1 & z_2 \end{pmatrix} = 1z_2 - 1z_1 = z_2 - z_1.$$

- ok

$$\begin{pmatrix} 1 & z_1 - 1z_1 & z_1^2 - z_1z_1 & \dots & z_1^{n-1} - z_1z_1^{n-2} \\ 1 & z_2 - 1z_1 & z_2^2 - z_1z_2 & \dots & z_2^{n-1} - z_1z_2^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & z_n - 1z_1 & z_n^2 - z_1z_n & \dots & z_n^{n-1} - z_1z_n^{n-2} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & z_2 - z_1 & z_2^2 - z_1z_2 & \dots & z_2^{n-1} - z_1z_2^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & z_n - z_1 & z_n^2 - z_1z_n & \dots & z_n^{n-1} - z_1z_n^{n-2} \end{pmatrix}.$$

- you can just use cofactor expansion and because the top row is all 0s you immediately have

$$\det V' = \det \begin{pmatrix} z_2 - z_1 & z_2^2 - z_1z_2 & \dots & z_2^{n-1} - z_1z_2^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ z_n - z_1 & z_n^2 - z_1z_n & \dots & z_n^{n-1} - z_1z_n^{n-2} \end{pmatrix}.$$

- i dont feel like finishing this problem

Problem 2

Since the Hermitian conjugate is the complex conjugate and the transpose, both of which are linear, the Hermitian conjugate must also be linear. Since A, B are Hermitian, we have

$$(A + B)^\dagger = A^\dagger + B^\dagger = A + B.$$

We also have that $(AB)^\dagger = B^\dagger A^\dagger$ from properties of the transpose, and the fact that the complex conjugate distributes over multiplication. It follows that

$$(AB + BA)^\dagger = (AB)^\dagger + (BA)^\dagger = B^\dagger A^\dagger + A^\dagger B^\dagger = BA + AB = AB + BA.$$

Finally,

$$(i[A, B])^\dagger = (iAB - iBA)^\dagger = i^* B^\dagger A^\dagger - i^* A^\dagger B^\dagger = -iBA + iAB = i(AB - BA) = i[A, B].$$

Problem 3

Let $S = \{\vec{v}_i\}_{i \in I} \subseteq V$ be a linearly independent subset of a $\mathbb{C}$ -vector space V . First, suppose that for any $\vec{v}_i \in S$, there exists no linear combination in $S \setminus \{\vec{v}_i\}$ that equals to $\vec{v}_i$. Let $A = \{a_i\}_{i \in I} \subseteq \mathbb{C}$ such that

$$\sum_{i \in I} a_i \vec{v}_i = 0.$$

Suppose for contradiction that there exists at least one $a_j \in A$ that is nonzero. We have

$$\sum_{i \in I \setminus \{j\}} a_i \vec{v}_i = -a_j \vec{v}_j.$$

Because $\mathbb{C}$ is a field and since $-a_j \neq 0$, we have that for any $a_i \in A$, the number $-a_i/a_j \in \mathbb{C}$ is also a scalar, and thus

$$\sum_{i \in I \setminus \{j\}} -\frac{a_i}{a_j} \vec{v}_i = \vec{v}_j$$

is a linear combination in S that equals a different element of S , a contradiction. Therefore, we must have $a_i = 0$ for all $i \in I$.

For the converse, suppose the only collection $\{a_i\}_{i \in I} \subseteq \mathbb{C}$ solving the equation

$$\sum_{i \in I} a_i \vec{v}_i = 0$$

is $a_i = 0$ for all i . Suppose for contradiction that S is not linearly independent; that is, there exists at least one $\vec{v}_j \in S$ such that there exists at least one collection $\{a_i\}_{i \in I \setminus \{j\}} \subseteq \mathbb{C}$ such that

$$\sum_{i \in I \setminus \{j\}} a_i \vec{v}_i = \vec{v}_j.$$

If we define $a_j = 1$, we can write

$$\sum_{i \in I \setminus \{j\}} a_i \vec{v}_i = a_j \vec{v}_j.$$

By subtracting $a_j \vec{v}_j$ from both sides, we have

$$\sum_{i \in I} a_i \vec{v}_i = 0.$$

However, we have $a_j = 1 \neq 0$, and thus there exists $j \in I$ such that $a_j \neq 0$, a contradiction. Therefore, S is linearly independent.

Problem 4

Let the vector $\mathbf{b}_i$ represent the i th column of B . If $\det B \neq 0$, then B is invertible, and thus

$$B\mathbf{v} = 0 \iff \mathbf{v} = B^{-1}0 = 0.$$

If v^i is the i th component of $\mathbf{v}$, we then have

$$v^i \mathbf{b}_i = 0 \iff \mathbf{v} = 0,$$

using Einstein's notation. Therefore, the columns of B are linearly independent.

Conversely, if $\mathbf{b}_i$ are not linearly independent, then there exists a nonzero collection of scalars $\{v^i\}$ such that $v^i \mathbf{b}_i = 0$. We then have

$$v^i \mathbf{b}_i = B\mathbf{v} = 0$$

for $\mathbf{v} \neq 0$, and thus B is not invertible, and thus $\det B = 0$.